

FOR EQUATIONS SEE NATIONAL DESIGN STANDARD

``SPAN OF FLOOR JOIST`` EXAMPLE

Southern Yellow Pine Reference Design Values:

wood = No 1 Standard Southern Pine

snow load = 55 lb/sqft for Acton MA

load duration factor (CD) = 1.6 for ten minutes

Possible Values: 1.6 for ten minutes
 1.25 for seven days
 1.15 for two months (snow load)
 0.9 for dead load

deflection limit = span/360

design bearing length = 1.5 inches

TO GET ALLOWED WEIGHT PER SQUARE FOOT,
 TAKE ALLOWED WEIGHT FOR ONE STRINGER PER FOOT,
 MULITPLY BY THE NUMBER OF STRINGERS,
 AND DIVIDE BY THE LENGTH OF THE TREAD IN FEET.

- (1) ALLOWED WEIGHT IN LBF/FT BY MOMENT CAPACITY
- (2) ALLOWED WEIGHT IN LBF/FT BY SHEAR
- (3) ALLOWED WEIGHT IN LBF/FT BY DEFLECTION
- (4) ALLOWED WEIGHT IN LBF/FT FOR 1.5 INCH BEARING

Allowable

Weight Is Proportional To

- | | ----- |
|-----|--|
| (1) | stringer width, CD, $1/\text{span}^2$ |
| (2) | stringer width, CD, $1/\text{span}$ |
| (3) | stringer width, deflection limit/span, $1/\text{span}^3$ |
| (4) | stringer width, bearing length, $1/\text{span}$ |

- NOTES: (1) LRDF for pedestrian bridges requires
 90 lbf / sqft, deflection limit = span/360
- (2) Bearing should be increased to
 2 inches for 2x10's
 2.5 inches for 2x12's
- (3) Loads (1) and (2) must be multiplied by
 $\text{CD-two-months}/\text{CD-10-minutes} = 1.15/1.6 = 71\%$
 to accommodate snow load. For 55 lb/sqft
 with 4 ft treads and 2 stringers, this load
 per string is 110 lb/ft = $71\% * 155 \text{ lb/ft}$.
- (4) For two-stringer boardwalk sections with
 3 ft or 4 ft treads, this is met by
 2x4's for 4 ft span **
 2x6's for 6 ft span
 2x8's for 8 ft span
 2x10's for 10 ft span
 2x12's for 12 ft span
 4x4's for 5 ft span
 4x6's for 8 ft span
 4x8's for 10 ft span
 4x10's for 13.5 ft span
 4x12's for 16 ft span
 (** but NOT for 4 ft treads)
- (5) If you add a middle stringer (to 3 total)
 you can increase the span by a factor of
 the cube root of 1.5 = 1.14.
- (6) If you double the number of stringers (to 4)
 you can increase the span by a factor of
 the cube root of 2 = 1.26.
- (7) If you triple the number of stringers (to 6)
 you can increase the span by a factor of
 the cube root of 3 = 1.44.

SPAN FT	2x4	2x6	2x8	2x10	2x12	SPAN FT	2x4	2x6	2x8	2x10	2x12
4.0 (1)	260.31	578.53	930.79	1497.34	2109.38	10.0 (1)	41.65	92.56	148.93	239.57	337.50
(2)	475.30	746.90	984.55	1256.15	1527.75	(2)	190.12	298.76	393.82	502.46	611.10
(3)	178.65	693.23	1587.83	3297.72	5932.62	(3)	11.43	44.37	101.62	211.05	379.69
(4)	425.87	425.87	425.87	425.87	425.87	(4)	170.35	170.35	170.35	170.35	170.35
5.0 (1)	166.60	370.26	595.71	958.30	1350.00	11.0 (1)	34.42	76.50	123.08	198.00	278.93
(2)	380.24	597.52	787.64	1004.92	1222.20	(2)	172.84	271.60	358.02	456.78	555.55
(3)	91.47	354.93	812.97	1688.43	3037.50	(3)	8.59	33.33	76.35	158.57	285.26
(4)	340.70	340.70	340.70	340.70	340.70	(4)	154.86	154.86	154.86	154.86	154.86
6.0 (1)	115.69	257.12	413.69	665.49	937.50	12.0 (1)	28.92	64.28	103.42	166.37	234.38
(2)	316.87	497.93	656.37	837.43	1018.50	(2)	158.43	248.97	328.18	418.72	509.25
(3)	52.93	205.40	470.47	977.10	1757.81	(3)	6.62	25.68	58.81	122.14	219.73
(4)	283.91	283.91	283.91	283.91	283.91	(4)	141.96	141.96	141.96	141.96	141.96
7.0 (1)	85.00	188.91	303.93	488.93	688.78	13.0 (1)	24.64	54.77	88.12	141.76	199.70
(2)	271.60	426.80	562.60	717.80	873.00	(2)	146.25	229.82	302.94	386.51	470.08
(3)	33.33	129.35	296.27	615.32	1106.96	(3)	5.20	20.19	46.25	96.06	172.82
(4)	243.35	243.35	243.35	243.35	243.35	(4)	131.04	131.04	131.04	131.04	131.04
8.0 (1)	65.08	144.63	232.70	374.34	527.34	14.0 (1)	21.25	47.23	75.98	122.23	172.19
(2)	237.65	373.45	492.28	628.08	763.88	(2)	135.80	213.40	281.30	358.90	436.50
(3)	22.33	86.65	198.48	412.22	741.58	(3)	4.17	16.17	37.03	76.91	138.37
(4)	212.93	212.93	212.93	212.93	212.93	(4)	121.68	121.68	121.68	121.68	121.68
9.0 (1)	51.42	114.28	183.86	295.77	416.67	15.0 (1)	18.51	41.14	66.19	106.48	150.00
(2)	211.24	331.96	437.58	558.29	679.00	(2)	126.75	199.17	262.55	334.97	407.40
(3)	15.68	60.86	139.40	289.51	520.83	(3)	3.39	13.15	30.11	62.53	112.50
(4)	189.28	189.28	189.28	189.28	189.28	(4)	113.57	113.57	113.57	113.57	113.57

SPAN FT	2x4	2x6	2x8	2x10	2x12	SPAN FT	2x4	2x6	2x8	2x10	2x12
16.0 (1)	16.27	36.16	58.17	93.58	131.84	22.0 (1)	8.61	19.12	30.77	49.50	69.73
(2)	118.83	186.73	246.14	314.04	381.94	(2)	86.42	135.80	179.01	228.39	277.77
(3)	2.79	10.83	24.81	51.53	92.70	(3)	1.07	4.17	9.54	19.82	35.66
(4)	106.47	106.47	106.47	106.47	106.47	(4)	77.43	77.43	77.43	77.43	77.43
17.0 (1)	14.41	32.03	51.53	82.90	116.78	23.0 (1)	7.87	17.50	28.15	45.29	63.80
(2)	111.84	175.74	231.66	295.56	359.47	(2)	82.66	129.90	171.23	218.46	265.70
(3)	2.33	9.03	20.68	42.96	77.28	(3)	0.94	3.65	8.35	17.35	31.21
(4)	100.20	100.20	100.20	100.20	100.20	(4)	74.06	74.06	74.06	74.06	74.06
18.0 (1)	12.85	28.57	45.97	73.94	104.17	24.0 (1)	7.23	16.07	25.86	41.59	58.59
(2)	105.62	165.98	218.79	279.14	339.50	(2)	79.22	124.48	164.09	209.36	254.62
(3)	1.96	7.61	17.42	36.19	65.10	(3)	0.83	3.21	7.35	15.27	27.47
(4)	94.64	94.64	94.64	94.64	94.64	(4)	70.98	70.98	70.98	70.98	70.98
19.0 (1)	11.54	25.64	41.25	66.36	93.49	25.0 (1)	6.66	14.81	23.83	38.33	54.00
(2)	100.06	157.24	207.27	264.45	321.63	(2)	76.05	119.50	157.53	200.98	244.44
(3)	1.67	6.47	14.82	30.77	55.36	(3)	0.73	2.84	6.50	13.51	24.30
(4)	89.66	89.66	89.66	89.66	89.66	(4)	68.14	68.14	68.14	68.14	68.14
20.0 (1)	10.41	23.14	37.23	59.89	84.38	26.0 (1)	6.16	13.69	22.03	35.44	49.93
(2)	95.06	149.38	196.91	251.23	305.55	(2)	73.12	114.91	151.47	193.25	235.04
(3)	1.43	5.55	12.70	26.38	47.46	(3)	0.65	2.52	5.78	12.01	21.60
(4)	85.17	85.17	85.17	85.17	85.17	(4)	65.52	65.52	65.52	65.52	65.52
21.0 (1)	9.44	20.99	33.77	54.33	76.53	27.0 (1)	5.71	12.70	20.43	32.86	46.30
(2)	90.53	142.27	187.53	239.27	291.00	(2)	70.41	110.65	145.86	186.10	226.33
(3)	1.23	4.79	10.97	22.79	41.00	(3)	0.58	2.25	5.16	10.72	19.29
(4)	81.12	81.12	81.12	81.12	81.12	(4)	63.09	63.09	63.09	63.09	63.09

SPAN FT	4x4	4x6	4x8	4x10	4x12	SPAN FT	4x4	4x6	4x8	4x10	4x12
4.0 (1)	607.40	1349.91	2171.85	3493.80	4921.88	10.0 (1)	97.18	215.98	347.50	559.01	787.50
(2)	1109.03	1742.77	2297.28	2931.02	3564.75	(2)	443.61	697.11	918.91	1172.41	1425.90
(3)	416.84	1617.53	3704.93	7694.68	13842.77	(3)	26.68	103.52	237.12	492.46	885.94
(4)	993.69	993.69	993.69	993.69	993.69	(4)	397.48	397.48	397.48	397.48	397.48
5.0 (1)	388.73	863.94	1389.99	2236.03	3150.00	11.0 (1)	80.32	178.50	287.19	461.99	650.83
(2)	887.23	1394.21	1837.83	2344.81	2851.80	(2)	403.28	633.73	835.38	1065.82	1296.27
(3)	213.42	828.18	1896.92	3939.68	7087.50	(3)	20.04	77.78	178.15	369.99	665.62
(4)	794.96	794.96	794.96	794.96	794.96	(4)	361.34	361.34	361.34	361.34	361.34
6.0 (1)	269.95	599.96	965.27	1552.80	2187.50	12.0 (1)	67.49	149.99	241.32	388.20	546.88
(2)	739.36	1161.84	1531.52	1954.01	2376.50	(2)	369.68	580.92	765.76	977.01	1188.25
(3)	123.51	479.27	1097.76	2279.91	4101.56	(3)	15.44	59.91	137.22	284.99	512.70
(4)	662.46	662.46	662.46	662.46	662.46	(4)	331.23	331.23	331.23	331.23	331.23
7.0 (1)	198.33	440.79	709.18	1140.83	1607.14	13.0 (1)	57.50	127.80	205.62	330.77	465.98
(2)	633.73	995.87	1312.73	1674.87	2037.00	(2)	341.24	536.24	706.86	901.85	1096.85
(3)	77.78	301.81	691.30	1435.74	2582.91	(3)	12.14	47.12	107.93	224.15	403.25
(4)	567.83	567.83	567.83	567.83	567.83	(4)	305.75	305.75	305.75	305.75	305.75
8.0 (1)	151.85	337.48	542.96	873.45	1230.47	14.0 (1)	49.58	110.20	177.29	285.21	401.79
(2)	554.52	871.38	1148.64	1465.51	1782.38	(2)	316.87	497.93	656.37	837.43	1018.50
(3)	52.11	202.19	463.12	961.84	1730.35	(3)	9.72	37.73	86.41	179.47	322.86
(4)	496.85	496.85	496.85	496.85	496.85	(4)	283.91	283.91	283.91	283.91	283.91
9.0 (1)	119.98	266.65	429.01	690.13	972.22	15.0 (1)	43.19	95.99	154.44	248.45	350.00
(2)	492.90	774.56	1021.01	1302.67	1584.33	(2)	295.74	464.74	612.61	781.60	950.60
(3)	36.60	142.01	325.26	675.53	1215.28	(3)	7.90	30.67	70.26	145.91	262.50
(4)	441.64	441.64	441.64	441.64	441.64	(4)	264.99	264.99	264.99	264.99	264.99

SPAN FT	4x4	4x6	4x8	4x10	4x12	SPAN FT	4x4	4x6	4x8	4x10	4x12
16.0 (1)	37.96	84.37	135.74	218.36	307.62	22.0 (1)	20.08	44.62	71.80	115.50	162.71
(2)	277.26	435.69	574.32	732.75	891.19	(2)	201.64	316.87	417.69	532.91	648.14
(3)	6.51	25.27	57.89	120.23	216.29	(3)	2.51	9.72	22.27	46.25	83.20
(4)	248.42	248.42	248.42	248.42	248.42	(4)	180.67	180.67	180.67	180.67	180.67
17.0 (1)	33.63	74.74	120.24	193.43	272.49	23.0 (1)	18.37	40.83	65.69	105.67	148.87
(2)	260.95	410.06	540.54	689.65	838.76	(2)	192.88	303.09	399.53	509.74	619.96
(3)	5.43	21.07	48.26	100.24	180.33	(3)	2.19	8.51	19.49	40.48	72.81
(4)	233.81	233.81	233.81	233.81	233.81	(4)	172.82	172.82	172.82	172.82	172.82
18.0 (1)	29.99	66.66	107.25	172.53	243.06	24.0 (1)	16.87	37.50	60.33	97.05	136.72
(2)	246.45	387.28	510.51	651.34	792.17	(2)	184.84	290.46	382.88	488.50	594.13
(3)	4.57	17.75	40.66	84.44	151.91	(3)	1.93	7.49	17.15	35.62	64.09
(4)	220.82	220.82	220.82	220.82	220.82	(4)	165.62	165.62	165.62	165.62	165.62
19.0 (1)	26.92	59.83	96.26	154.85	218.14	25.0 (1)	15.55	34.56	55.60	89.44	126.00
(2)	233.48	366.90	483.64	617.06	750.47	(2)	177.45	278.84	367.57	468.96	570.36
(3)	3.89	15.09	34.57	71.80	129.16	(3)	1.71	6.63	15.18	31.52	56.70
(4)	209.20	209.20	209.20	209.20	209.20	(4)	158.99	158.99	158.99	158.99	158.99
20.0 (1)	24.30	54.00	86.87	139.75	196.88	26.0 (1)	14.38	31.95	51.40	82.69	116.49
(2)	221.81	348.55	459.46	586.20	712.95	(2)	170.62	268.12	353.43	450.93	548.42
(3)	3.33	12.94	29.64	61.56	110.74	(3)	1.52	5.89	13.49	28.02	50.41
(4)	198.74	198.74	198.74	198.74	198.74	(4)	152.88	152.88	152.88	152.88	152.88
21.0 (1)	22.04	48.98	78.80	126.76	178.57	27.0 (1)	13.33	29.63	47.67	76.68	108.02
(2)	211.24	331.96	437.58	558.29	679.00	(2)	164.30	258.19	340.34	434.22	528.11
(3)	2.88	11.18	25.60	53.18	95.66	(3)	1.36	5.26	12.05	25.02	45.01
(4)	189.28	189.28	189.28	189.28	189.28	(4)	147.21	147.21	147.21	147.21	147.21

SPAN FT	4x2	6x2	8x2	10x2	12x2	SPAN FT	6x4	8x4	10x4	12x4
2.0 (1)	490.88	725.79	885.86	1165.50	1350.00	2.0 (1)	3607.93	4403.62	5816.71	6737.50
(2)	950.60	1493.80	1969.10	2512.30	3055.50	(2)	3485.53	4594.57	5862.03	7129.50
(3)	262.50	412.50	543.75	693.75	843.75	(3)	5240.28	6907.64	8813.19	10718.75
(4)	1987.39	3123.04	4116.73	5252.38	6388.03	(4)	3123.04	4116.73	5252.38	6388.03
2.5 (1)	314.16	464.51	566.95	745.92	864.00	2.5 (1)	2309.08	2818.32	3722.69	4312.00
(2)	760.48	1195.04	1575.28	2009.84	2444.40	(2)	2788.43	3675.65	4689.63	5703.60
(3)	134.40	211.20	278.40	355.20	432.00	(3)	2683.02	3536.71	4512.36	5488.00
(4)	1589.91	2498.43	3293.39	4201.91	5110.43	(4)	2498.43	3293.39	4201.91	5110.43
3.0 (1)	218.17	322.57	393.72	518.00	600.00	3.0 (1)	1603.53	1957.16	2585.20	2994.44
(2)	633.73	995.87	1312.73	1674.87	2037.00	(2)	2323.69	3063.04	3908.02	4753.00
(3)	77.78	122.22	161.11	205.56	250.00	(3)	1552.67	2046.71	2611.32	3175.93
(4)	1324.93	2082.03	2744.49	3501.59	4258.69	(4)	2082.03	2744.49	3501.59	4258.69
3.5 (1)	160.29	236.99	289.26	380.57	440.82	8.0 (1)	225.50	275.23	363.54	421.09
(2)	543.20	853.60	1125.20	1435.60	1746.00	(2)	871.38	1148.64	1465.51	1782.38
(3)	48.98	76.97	101.46	129.45	157.43	(3)	81.88	107.93	137.71	167.48
(4)	1135.65	1784.59	2352.42	3001.36	3650.30	(4)	780.76	1029.18	1313.10	1597.01
4.0 (1)	122.72	181.45	221.46	291.38	337.50	9.0 (1)	178.17	217.46	287.24	332.72
(2)	475.30	746.90	984.55	1256.15	1527.75	(2)	774.56	1021.01	1302.67	1584.33
(3)	32.81	51.56	67.97	86.72	105.47	(3)	57.51	75.80	96.72	117.63
(4)	993.69	1561.52	2058.37	2626.19	3194.02	(4)	694.01	914.83	1167.20	1419.56
						9.5 (1)	159.91	195.17	257.80	298.61
						(2)	733.80	967.28	1234.11	1500.95
						(3)	48.90	64.45	82.23	100.01
						(4)	657.48	866.68	1105.76	1344.85
						10.0 (1)	144.32	176.14	232.67	269.50
						(2)	697.11	918.91	1172.41	1425.90
						(3)	41.92	55.26	70.51	85.75
						(4)	624.61	823.35	1050.48	1277.61